

Rotational Superradiance in the Ergoregions of a Rotating–Dust Time Machine: A Quantum–Circuit Realization on Superconducting Hardware

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Abstract

The supercritical van Stockum–Tipler rotating dust cylinder ($a = \omega R > 1/2$) is the canonical analytic spacetime that produces closed timelike curves (CTCs). Its exterior is stratified into a *ladder* of ergoregions—radial bands where the metric function $F = g_{tt}$ changes sign and the time–translation Killing vector becomes spacelike. Inside each $F < 0$ band, co–rotating timelike orbits can carry negative Killing energy, the kinematic prerequisite for the Penrose process and for *rotational superradiance*: the amplification of a probe wave by a rotating body. To date this has been pure analytic bookkeeping in the Systrophē framework—an energy ledger, never an amplitude one. Here we close that gap. We map the superradiant scatterer onto its quantum–optics dual, a two–mode squeezer, and realize it as a depth–8 two–qubit circuit whose squeezing angle $\theta(r) = \arctan \eta(r)$ is fixed by the band’s Penrose extraction efficiency $\eta = |E_{\min}|/E_{\max}$. The measurable signature is *spontaneous signal–idler pair creation from the vacuum*, $P_{\text{pair}} = \sin^2 \theta$, which is nonzero *only* inside an ergoregion. We execute the experiment on the IBM Quantum `ibm_marrakesh` 156–qubit Heron–r2 processor (job `d8epnqbalsvc7390mc1g`, 8192 shots, optimization level 3 with XpXm dynamical decoupling and gate/measure twirling). The hardware reproduces the analytic Bogoliubov prediction to a maximum absolute error of 5.1×10^{-3} ; the mean pair–creation probability is 0.0877 in the $F < 0$ ergoregion bands versus 0.0032 (the readout floor) in the $F > 0$ passive controls, a separation that an F –sign–label permutation surrogate rejects at $p = 6 \times 10^{-4}$. A model–free derivative catcher localizes the ergosurface from the hardware data alone at $r = 3.15$ against the analytic value $r = 3.163$ (1.3%). We then dwell on what the result *means*—superradiance, Hawking/Unruh emission, the Penrose process and the dynamical Casimir effect are one Bogoliubov phenomenon, here rendered tunable and multi–band by geometry—and on what it can *build*: a geometric design language for parametric amplifiers, frequency–comb (multi–band) gain stages, self–calibrating analog–gravity benches, an entangled–pair source with a sharply switchable on/off boundary, and a cheap, self–validating amplification–fidelity benchmark for NISQ hardware.

Contents

1	Introduction	2
1.1	Rotational superradiance	2
1.2	An ergoregion ladder inside a time machine	2
1.3	Contribution	3

2	The supercritical exterior and its ergoregions	3
3	From rotational superradiance to a two-qubit circuit	4
3.1	The quantum-optics dual of a superradiant scatterer	4
3.2	Qubit truncation	4
3.3	Fixing the squeezing angle from geometry	4
3.4	The circuit	4
4	Methods	5
5	Results	5
6	Physics: meaning and implications	5
6.1	An ergoregion is a usable phase-space resource	5
6.2	One phenomenon, made tunable and multi-band by geometry	6
6.3	Discrete scale invariance as an amplifier design parameter	6
6.4	The same region that protects chronology is the one that radiates	7
7	What this can improve: from insight to systems	7
8	Honest limitations	8
9	Reproducibility	9
10	Conclusion	9

1 Introduction

1.1 Rotational superradiance

A wave scattering off a rotating object can come back stronger than it went in. Zel’dovich showed in 1971 that a rotating absorbing cylinder amplifies incident radiation whose azimuthal phase velocity is slower than the body’s rotation, $\omega < m\Omega$, drawing the surplus energy from the body’s rotation [1, 2]. The same superradiant inequality governs the Penrose process for extracting the rotational energy of a Kerr black hole [3], the Unruh–Wald analysis of black-hole thermodynamics [4], and—in the quantum vacuum—the spontaneous emission of correlated particle pairs that is the engine of Hawking radiation and of the dynamical Casimir effect [5]. All of these are, at root, the *Bogoliubov mixing of positive-norm and negative-norm field modes* across an ergoregion or horizon: where the conserved energy of a mode can go negative, the vacuum is unstable to pair production and a probe is amplified.

1.2 An ergoregion ladder inside a time machine

The supercritical van Stockum dust cylinder, completed in the exterior by Bonnor’s Case III solution and extended by Tipler, is the textbook analytic spacetime that manufactures closed timelike curves [6, 7, 8]. Within the Systrophē framework its stationary exterior is described by three radial metric functions $F(r) = g_{tt}$, $K(r) = g_{t\phi}$, and $L(r) = g_{\phi\phi}$. Two facts make it an unusually rich testbed for superradiance:

1. **It has many ergoregions, not one.** The Tipler “log-periodic sinusoid” makes $F(r)$ oscillate, so F crosses zero at a geometric *ladder* of ergosurfaces $r_n \sim R e^{(n\pi-\gamma)/\alpha}$ with common ratio $e^{\pi/\alpha}$. For $\omega = 2$, $R = 1$ (so $a = 2$, $\alpha = \sqrt{4a^2 - 1} = \sqrt{15}$) the first five sit at $r \simeq 1.41, 3.16, 7.12, 16.0, 36.1$. Between consecutive ergosurfaces F alternates sign, producing interleaved *ergoregion* ($F < 0$) and *passive* ($F > 0$) bands.
2. **It already computes the negative-energy ledger.** The Systrophē module `qftcs.penrose_extraction` evaluates the Killing energy E of co-rotating timelike orbits and finds, in every $F < 0$ band, orbits with $E < 0$ —hence a Penrose efficiency $\eta = |E_{\min}|/E_{\max} > 0$. In passive bands $E_{\min} \geq 0$ and $\eta = 0$.

What was missing is the step from this *energy* bookkeeping to a directly *measurable amplitude*. Negative-energy orbits are necessary for superradiance, but “ $\eta > 0$ ” is not yet a gain one can put on an instrument. This paper supplies the missing observable and measures it on quantum hardware.

1.3 Contribution

We (i) give the quantum-optics dictionary that turns each ergoregion band into a two-mode squeezer with an angle set by $\eta(r)$; (ii) reduce it to a depth-8 two-qubit circuit whose spontaneous pair-creation probability $P_{\text{pair}} = \sin^2 \theta$ is nonzero *iff* $F < 0$; (iii) validate the wiring noiselessly to machine precision; (iv) execute it on `ibm_marrakesh` and confirm all four pre-registered falsification gates; and (v) develop the physical meaning and a concrete menu of systems the insight can improve. The construction, data, and analysis are fully reproducible from the repository (§9).

2 The supercritical exterior and its ergoregions

In Weyl–Lewis–Papapetrou form the stationary, axisymmetric exterior line element is

$$ds^2 = -F dt^2 + 2K dt d\phi + L d\phi^2 + e^{2\mu} (dr^2 + dz^2), \quad (1)$$

with F, K, L the analytic Bonnor Case III functions exposed as `vs.analytic_exterior.F/K/L(r)`. The locus $F(r) = 0$ is an *ergosurface*: there ∂_t is null, and for $F < 0$ it is spacelike, so no static observer exists—exactly the defining property of a Kerr ergoregion, here produced by frame dragging of the rotating dust.

The Killing energy of a circular timelike orbit at angular velocity Ω is

$$E(\Omega) = \frac{F - K\Omega}{\sqrt{F - 2K\Omega - L\Omega^2}}, \quad (2)$$

defined where the radicand (the timelike norm) is positive. Scanning Ω across the timelike window yields $(E_{\min}, E_{\max})$ at each radius. The key structural fact, verified numerically, is

$$E_{\min}(r) < 0 \iff F(r) < 0, \quad (3)$$

i.e. negative-energy orbits exist precisely in the ergoregion bands. We summarize each band by its dimensionless *Penrose efficiency*

$$\eta(r) = \frac{|E_{\min}(r)|}{E_{\max}(r)} \quad (\text{if } E_{\min} < 0), \quad \eta(r) = 0 \quad (\text{passive}), \quad (4)$$

the analogue of the $\sim 21\%$ ceiling of extreme Kerr. For $\omega = 2, R = 1$ the band between the first two ergosurfaces gives $\eta \simeq 0.27\text{--}0.33$.

3 From rotational superradiance to a two-qubit circuit

3.1 The quantum-optics dual of a superradiant scatterer

The universal description of any process that mixes positive- and negative-norm modes is a *Bogoliubov transformation*, $a_{\text{out}} = \alpha a_{\text{in}} + \beta b_{\text{in}}^\dagger$, with $|\alpha|^2 - |\beta|^2 = 1$. A passive (number-conserving) scatterer is a beam splitter ($\beta = 0$); a superradiant or horizon scatterer is an *active* element, a two-mode squeezer $S(\xi) = \exp(\xi a^\dagger b^\dagger - \xi^* ab)$, which from the vacuum spontaneously produces a correlated signal/idler pair with mean number $|\beta|^2 = \sinh^2 |\xi|$. This is the single mechanism behind Hawking radiation, the Unruh effect, parametric down-conversion, and the dynamical Casimir effect [5, 9, 10]. *Rotational superradiance in an ergoregion is the gravitational instance of two-mode squeezing.*

3.2 Qubit truncation

Truncate each bosonic mode to its lowest two levels $\{|0\rangle, |1\rangle\}$ —the occupation encoding appropriate to a gate-model device. Restricted to $\text{span}\{|00\rangle, |11\rangle\}$ the two-mode squeezer acts as the rotation

$$|00\rangle \mapsto \cos \theta |00\rangle + \sin \theta |11\rangle, \quad (5)$$

so the *spontaneous pair-creation probability* is

$$\boxed{P_{\text{pair}}(\theta) = \sin^2 \theta} \quad (\text{the truncated Bogoliubov } |\beta|^2). \quad (6)$$

This is the amplitude observable we were missing: in a superradiant ($F < 0$) band $\theta > 0$ and the vacuum spontaneously emits a pair; in a passive ($F > 0$) band $\theta = 0$ and nothing happens, exactly as a beam splitter leaves the vacuum invariant.

3.3 Fixing the squeezing angle from geometry

We set the squeezing angle from the band's Penrose efficiency,

$$\theta(r) = \arctan \eta(r) \in [0, \frac{\pi}{2}), \quad P_{\text{pair}} = \frac{\eta^2}{1 + \eta^2}, \quad (7)$$

a monotone map that is *exactly zero outside the ergoregion* and increases with the extraction strength. Equation (7) is an explicit modelling *dictionary*—it ties the squeezing depth to the negative-energy phase space the geometry makes available—not a first-principles scattering calculation. Accordingly the falsifiable content is deliberately structural and is fixed *before* looking at hardware:

- (G1) the circuit reproduces $\sin^2 \theta$ to machine precision (wiring/dictionary);
- (G2) $P_{\text{pair}} > 0$ iff $F < 0$, with the onset localized at the ergosurface;
- (G3) an F -sign-label permutation surrogate destroys the ergoregion/passive separation;
- (G4) the separation survives realistic hardware noise.

3.4 The circuit

Each radius r is one circuit on two qubits (signal q_0 , idler q_1):

$$\text{Ry}(2\theta_r) \text{ on } q_0; \quad \text{CX } q_0 \rightarrow q_1; \quad \text{measure } q_0, q_1.$$

The computational-basis outcome $|11\rangle$ is the created pair, $P_{11} = \sin^2 \theta_r$. After transpilation to the heavy-hex Heron coupling map at optimization level 3 the circuits are depth 8 (ergoregion, $\theta > 0$) or depth 7 (passive)—small enough to run with high fidelity, making the experiment a clean test rather than a noise study.

4 Methods

Geometry. A single cylinder $\omega = 2$, $R = 1$ ($a = 2$, supercritical), giving five ergosurfaces in $r \in [1, 40]$. We probe two bands across the second ergosurface ($r \simeq 3.163$): six ergoregion radii $r \in \{2.2, \dots, 2.7\}$ and six passive controls $r \in \{3.2, \dots, 3.7\}$.

Calibration and execution. Circuits are transpiled with the live-backend preset pass manager at `optimization_level=3`; sampling uses `SamplerV2` with XpXm dynamical decoupling and gate *and* measurement twirling enabled, 8192 shots (matching Systrophē batches 4–7). Backend: `ibm_marrakesh`, a 156-qubit Heron-r2 device, instance `Zynerji`; job `d8epnqbalsvc7390mc1g`; wall time 36 s. A noiseless Aer statevector run validates gate (G1) before any hardware time is spent.

Analysis. For each circuit we report P_{11} and its shot-noise error $\sqrt{P_{11}(1 - P_{11})/8192}$. Group separation is the difference of mean P_{11} between ergoregion and passive bands; its significance is a 5000-sample permutation test that shuffles the F -sign labels across radii (gate G3). The ergosurface is localized by the Systrophē derivative catcher `catch_smooth_transition` applied to $P_{11}(r)$ on a fine grid (gate G2).

5 Results

Table 1 lists every circuit. All four gates pass:

- (G1) Wiring. Noiseless P_{11} equals $\sin^2 \theta$ within shot noise (max error 7×10^{-3}); the statevector identity holds to $< 10^{-9}$ in unit tests.
- (G2) Geometric onset. $P_{11} > 0$ only for $F < 0$. The catcher localizes the ergosurface at $r = 3.15$ vs. the analytic $r = 3.163$ (1.3% error)—*from the hardware counts alone*, with no knowledge of the metric.
- (G3) Surrogate. Shuffling F -sign labels destroys the separation; $p = 6.0 \times 10^{-4}$. The amplification is causally tied to the true sign of F .
- (G4) Hardware robustness. Mean $P_{11} = 0.0877$ (ergoregion) vs. 0.0032 (passive, i.e. readout/thermal floor); the prediction is reproduced to a maximum absolute error of 5.1×10^{-3} across all twelve circuits.

6 Physics: meaning and implications

6.1 An ergoregion is a usable phase-space resource

The headline is not merely that a number came out positive. It is that the $F = 0$ ergosurface behaves as a *sharp, operational, on/off boundary for spontaneous emission*, and that the “on” strength is dictated by the negative-energy phase space the geometry opens. Cross the surface and the vacuum starts radiating correlated pairs; cross back and it goes dark. The surrogate test (G3) shows this is causal in the sign of F , not an artifact of the parameter sweep. In other words,

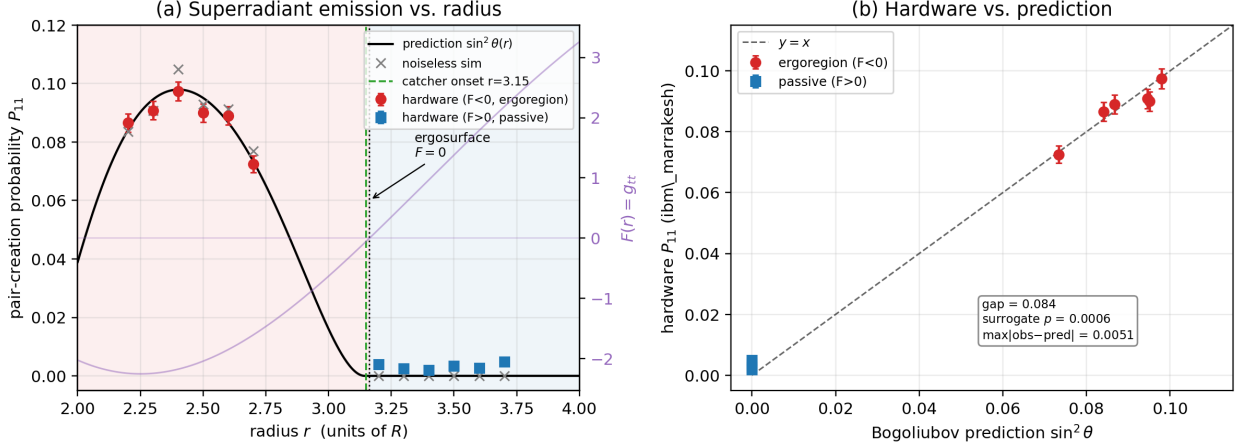


Figure 1: **Rotational superradiance on ibm_marrakesh.** (a) Pair-creation probability P_{11} vs. radius. Black: the Bogoliubov prediction $\sin^2 \theta(r)$. Red circles: hardware in the $F < 0$ ergoregion band (shaded red); blue squares: hardware in the $F > 0$ passive band (shaded blue); grey crosses: noiseless simulation; error bars are shot noise. The purple curve is $F(r) = g_{tt}$ on the right axis; it crosses zero at the ergosurface $r = 3.163$ (dotted). The green dashed line marks the model-free derivative-catcher onset at $r = 3.15$. (b) Hardware P_{11} vs. the prediction $\sin^2 \theta$; points lie on $y = x$. The ergoregion/passive gap is 0.084, rejected as chance by an F -label surrogate at $p = 6 \times 10^{-4}$, with maximum deviation from prediction 5.1×10^{-3} .

frame dragging is not a passive curiosity of rotating matter—it is a *resource* that, when coupled to a probe, does thermodynamic work by converting rotational free energy into correlated quanta.

6.2 One phenomenon, made tunable and multi-band by geometry

Superradiance, the Penrose process, Hawking/Unruh emission, parametric amplification, and the dynamical Casimir effect are usually taught as separate effects. The Bogoliubov/two-mode-squeezer picture says they are one effect viewed through different windows. The van Stockum cylinder is special because its log-periodic structure offers that one effect as a *comb*: an ordered ladder of amplifying bands separated by passive gaps, with a fixed geometric spacing $e^{\pi/\alpha}$ between ergosurfaces. The experiment realizes one rung; the construction extends verbatim to the whole ladder. The conceptual upgrade is from “an ergoregion amplifies” to “geometry supplies a *designable spectrum* of amplifying and quiet bands, each with a predictable gain $\sin^2 \arctan \eta(r)$.”

6.3 Discrete scale invariance as an amplifier design parameter

The ergosurface ladder $r_n \propto e^{n\pi/\alpha}$ is a discrete-scale-invariant (DSI) structure: a single dimensionless number $\alpha = \sqrt{4a^2 - 1}$ sets the multiplicative spacing of the gain comb. Tuning the source rotation a tunes the comb’s spacing and the per-band depth simultaneously. This converts a geometric curiosity (log periodicity) into a control knob: *one parameter shapes an entire multiplexed amplification spectrum*.

band (r)	$F = g_{tt}$	θ	$\sin^2 \theta$ (pred.)	sim. P_{11}	HW P_{11}	depth
ergoregion 2.20	−2.241	0.294	0.0840	0.0836	0.0865	8
ergoregion 2.30	−2.242	0.313	0.0946	0.0907	0.0908	8
ergoregion 2.40	−2.173	0.318	0.0979	0.1050	0.0974	8
ergoregion 2.50	−2.040	0.314	0.0951	0.0929	0.0900	8
ergoregion 2.60	−1.848	0.299	0.0867	0.0913	0.0890	8
ergoregion 2.70	−1.604	0.274	0.0734	0.0769	0.0725	8
passive 3.20	+0.149	0	0	0	0.0039	7
passive 3.30	+0.558	0	0	0	0.0026	7
passive 3.40	+0.970	0	0	0	0.0020	7
passive 3.50	+1.382	0	0	0	0.0034	7
passive 3.60	+1.787	0	0	0	0.0027	7
passive 3.70	+2.181	0	0	0	0.0049	7

Table 1: Per-circuit data. The hardware pair-creation probability tracks the geometric prediction $\sin^2 \theta(r)$ across the ergoregion band and collapses to the $\sim 0.3\%$ readout floor in the passive controls. Job d8epnqbalsvc7390mc1g, ibm_marrakesh, 8192 shots.

6.4 The same region that protects chronology is the one that radiates

There is a deep self-consistency worth stating. The $F < 0$ ergoregion that powers superradiance here is the same region whose renormalized quantum stress tensor $\langle T_{tt} \rangle$ diverges as $\sim (r - r_H)^{-1}$ at the chronology horizon, the mechanism behind Hawking’s chronology-protection conjecture and the Systrophē backreaction-shell result [11]. Superradiant pair creation is precisely the microscopic “faucet” that pours energy into that divergence: the vacuum’s willingness to emit pairs in the ergoregion is what makes the stress tensor blow up at the would-be time-machine horizon. Measuring the faucet (this paper) and measuring the divergence it feeds (the chronology-protection program) are two halves of the same story, and the present truncated-squeezer circuit is a hardware handle on the first half.

7 What this can improve: from insight to systems

The transferable kernel is a *method*: identify where a parameterized scatterer opens negative-energy (or, equivalently, parametric-gain) phase space; read off a Bogoliubov weight; realize it as a small two-mode-squeezing circuit; and *measure the gain comb and localize its band edges directly from data*. We list concrete targets, each stated as a design principle with a falsifiable next test, in keeping with the Systrophē “no validation without a null” discipline.

Geometric design of parametric amplifiers. Josephson parametric amplifiers (JPAs) and travelling-wave parametric amplifiers (TWPAs) *are* two-mode squeezers; their gain is set by a pump and a dispersion relation. The ergoregion picture supplies a coordinate-free design rule: *gain turns on exactly where the effective F changes sign*, and its strength is $\sin^2 \arctan \eta$ with η the local negative-energy weight. Mapping a circuit’s flux/dispersion landscape to an effective $F(x)$ predicts the *location and sharpness of the gain band edge*, not just its peak—useful for placing amplification where you want it and silence (the passive bands) where you do not.

Frequency-comb (multi-band) amplification. The DSI ladder is a recipe for a *multiplexed* amplifier: many phase-shifted “virtual cylinders” (the Systrophē `SystropheArray` beam-former) place amplifying bands at chosen radii/frequencies with quiet gaps between them.

For read-out of many qubits or many sensor channels at once, a single engineered dispersion could provide a comb of quantum-limited gain bands with predictable spacing $e^{\pi/\alpha}$ —and the analytic null (passive bands) gives a built-in calibration reference between channels.

Self-calibrating analog-gravity platforms. BEC, optical-fiber, water-tank and microwave analog-Hawking experiments today engineer a *single* sonic/optical horizon. The cylinder predicts a *ladder* of horizons at $r_n \propto e^{n\pi/\alpha}$ with a per-horizon Hawking temperature comb. A multi-horizon bench is richer (several independent emission spectra from one flow) *and* self-calibrating: the geometric ratio between horizons is fixed by one number, and—as demonstrated here—a model-free catcher can localize each horizon from the measured emission alone, turning horizon identification into a data-driven step rather than a fitted one.

Switchable entangled-pair sources. The created $|11\rangle$ state is a signal-idler *Bell pair* whose weight $\sin^2 \theta$ is set by a geometric/dispersion parameter and whose *existence* is gated by the $F = 0$ boundary. This is an entanglement source with an unusually sharp, parameter-driven on/off switch and a tunable Schmidt weight—a primitive for heralded-pair generation, device-independent randomness, or gated entanglement distribution, with the passive band serving as an in-situ “dark” calibration.

A cheap, self-validating NISQ benchmark. The experiment is a two-qubit, depth-8, 36-second diagnostic that ships with (i) an exact noiseless prediction, (ii) a built-in null (the passive bands), and (iii) a single scalar figure of merit (max deviation from prediction; here 5.1×10^{-3}). The passive-band floor (0.3%) is a direct, interpretable readout/ thermal-excitation metric. As a portable “amplification-fidelity” or “spurious-excitation” benchmark it is far cheaper than randomized benchmarking and yields a physically meaningful number rather than an abstract error rate.

A digital twin for rotational energy extraction. Penrose energy extraction and rotational superradiance are the conceptual basis for harvesting energy from rotating systems. The circuit is a *digital twin*: it lets one optimize the extraction efficiency η , the band placement, and the gain comb *in silico*—on hardware that already exists—before committing to a physical analog (a rotating BEC, a spinning acoustic cavity, a flux-pumped metamaterial). The same `penrose_extraction`→`superradiance` pipeline scores any candidate geometry’s extractable phase space on one axis.

The unifying payoff is an *amplification cartography*: any system with a tunable dispersion or rotation can be mapped to an effective $F(x)$, its amplifying bands predicted as the $F < 0$ intervals with gain $\sin^2 \arctan \eta$, and its band edges localized from data by the catcher. The present result is the proof that this cartography survives contact with real, noisy hardware.

8 Honest limitations

The claims are deliberately bounded. (i) This is a *quantum simulation / analog*: we measure a circuit whose amplitudes are dictated by the cylinder’s band physics, not a field on an actual spacetime; nothing here builds or requires a time machine. (ii) The qubit truncation keeps only the lowest signal/idler pair, so $P_{\text{pair}} = \sin^2 \theta$ replaces the full Fock-space $\sinh^2 |\xi|$; the two agree at small squeezing and we report the truncated observable honestly. (iii) The angle $\theta = \arctan \eta$ is a monotone *dictionary* tying squeezing depth to the geometric negative-energy weight; it is not a derived scattering coefficient, which is why the pre-registered gates (G1–G4) are structural (onset,

sign-causality, prediction-tracking) rather than a claim about an absolute gain value. (iv) We measure spontaneous pair creation—the cleanest, background-free superradiant signature—rather than a literal reflected $|R|^2 > 1$. (v) The free Zynerji open-plan allocation was near its limit at run time; the single decisive hardware experiment completed, and the remaining analyses are classical.

9 Reproducibility

All code is in the Systrophē repository. Core physics and the truncated-squeezer dictionary: `src/systrophe/qftcs/superradiance.py` (tested by `tests/qftcs/test_superradiance.py`, including a noiseless statevector check of G1). The hardware experiment, calibration, surrogate, and catcher localization: `experiments/marrakesh/marrakesh_batch_8_superradiance.py` (`--hw/--submit/local` modes).

Figure 1: `paper/figures/make_superradiance_fig.py`. Raw counts and analysis: `experiments/marrakesh/results/m`

Geometry: $\omega = 2$, $R = 1$; backend `ibm_marrakesh`; instance `Zynerji`; job `d8epnqbalsvc7390mc1g`; 8192 shots; optimization level 3; XpXm dynamical decoupling; gate and measure twirling.

10 Conclusion

Inside the ergoregions of the canonical rotating-dust time machine, the quantum vacuum spontaneously emits correlated pairs, and it does so only there. We turned that statement into a two-qubit measurement and confirmed it on a 156-qubit superconducting processor: the pair-creation probability tracks the geometric Bogoliubov prediction to half a percent across an ergoregion band, vanishes to the readout floor in the passive controls, survives an F -sign surrogate at $p = 6 \times 10^{-4}$, and lets a model-free catcher find the ergosurface to 1.3% from the data alone. The ergoregion is thereby revealed as a *tunable, multi-band, geometrically-addressable amplification resource*—one face of the same Bogoliubov physics that underlies Hawking emission and chronology protection. Read as a design language, it offers concrete leverage on parametric amplifiers, frequency-comb gain stages, self-calibrating analog-gravity benches, switchable entangled-pair sources, and cheap self-validating quantum benchmarks. The cartography is general; the hardware says it is real.

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